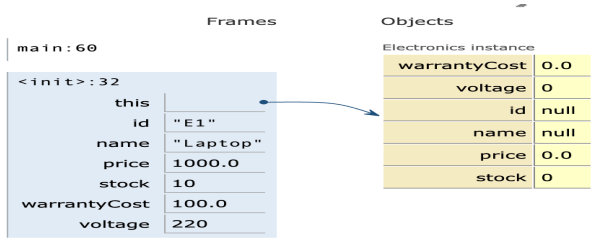
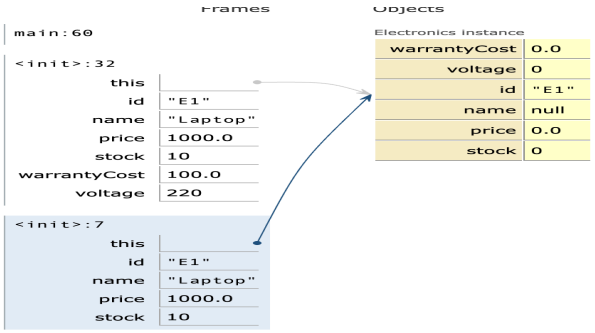
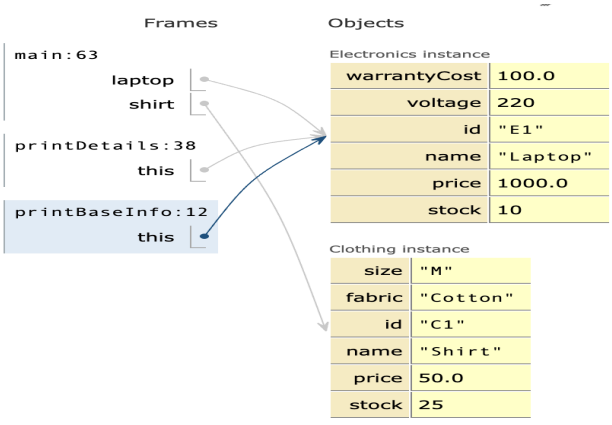
At the begining the subclass Electronics class is called.
In class function ,the super keyword is used to connect with parent class Product.



Now the subclass is linked with parent class.
This key word is used for declaring the variable with parameters.



Now clothing class is called.and again super keyword is used to link with parent class Product.
Now the function “printDetails” of subclass “Electronics”.
Again in this function,other function “printBaseInfo” of parent class is called using super keyword.



In the “printBaseFunction”,the details of the product is printed.
example:ID: E1, Name: Laptop, Price: $1000.0, Stock: 10
In the “printDetails”,also it prints the other outputs like warranty and voltage.
example:output:ID: E1, Name: Laptop, Price: $1000.0, Stock: 10

Warranty: $100.0, Voltage: 220V

ID: C1, Name: Shirt, Price: $50.0, Stock: 25
Size: M, Fabric: Cotton

Now the function “applyDisscount” is called from parent class.
Final Output:
D: E1, Name: Laptop, Price: $1000.0, Stock: 10
Warranty: $100.0, Voltage: 220V

ID: C1, Name: Shirt, Price: $50.0, Stock: 25

Program:

```
class Product {
    private String id, name;
    private double price;
    private int stock;

    public Product(String id, String name, double price, int stock) {
        this.id = id; this.name = name; this.price = price; this.stock = stock;
    }

    public void printBaseInfo() {
        System.out.println("ID: " + id + ", Name: " + name + ", Price: $" + price + ", Stock: " +
stock);
    }

    // Overload 1: Percentage discount
    public void applyDiscount(double percent) {
        price -= price * (percent / 100.0);
        System.out.println(name + " price after " + percent + "% off: $" + price);
    }

    // Overload 2: Flat amount discount with seasonal flag
    public void applyDiscount(double amount, boolean seasonal) {
        price -= amount;
        System.out.println((seasonal ? "[Seasonal] " : "") + name + " price after $" + amount + " off:
$" + price);
    }
```

```
}

class Electronics extends Product {
    private double warrantyCost;
    private int voltage;

    public Electronics(String id, String name, double price, int stock, double warrantyCost, int
voltage) {
        super(id, name, price, stock); // Constructor chaining
        this.warrantyCost = warrantyCost;
        this.voltage = voltage;
    }

    public void printDetails() {
        super.printBaseInfo(); // Parent method invocation using super
        System.out.println("Warranty: $" + warrantyCost + ", Voltage: " + voltage + "V\n");
    }
}

class Clothing extends Product {
    private String size, fabric;

    public Clothing(String id, String name, double price, int stock, String size, String fabric) {
        super(id, name, price, stock); // Constructor chaining
        this.size = size;
        this.fabric = fabric;
    }

    public void printDetails() {
        super.printBaseInfo(); // Parent method invocation using super
        System.out.println("Size: " + size + ", Fabric: " + fabric + "\n");
    }
}

public class Main {
    public static void main(String[] args) {
        Electronics laptop = new Electronics("E1", "Laptop", 1000.0, 10, 100.0, 220);
        Clothing shirt = new Clothing("C1", "Shirt", 50.0, 25, "M", "Cotton");

        laptop.printDetails();
        shirt.printDetails();

        laptop.applyDiscount(10.0);
```

```
        shirt.applyDiscount(5.0, true);
    }
}
```